

Reverse-Complement Aware Dilated CNN–BiGRU with Gated Attention for Regulatory DNA Prediction



EECS6895 Advanced Big Data & AI

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Tasks: ATAC-seq accessibility + CTCF
binding

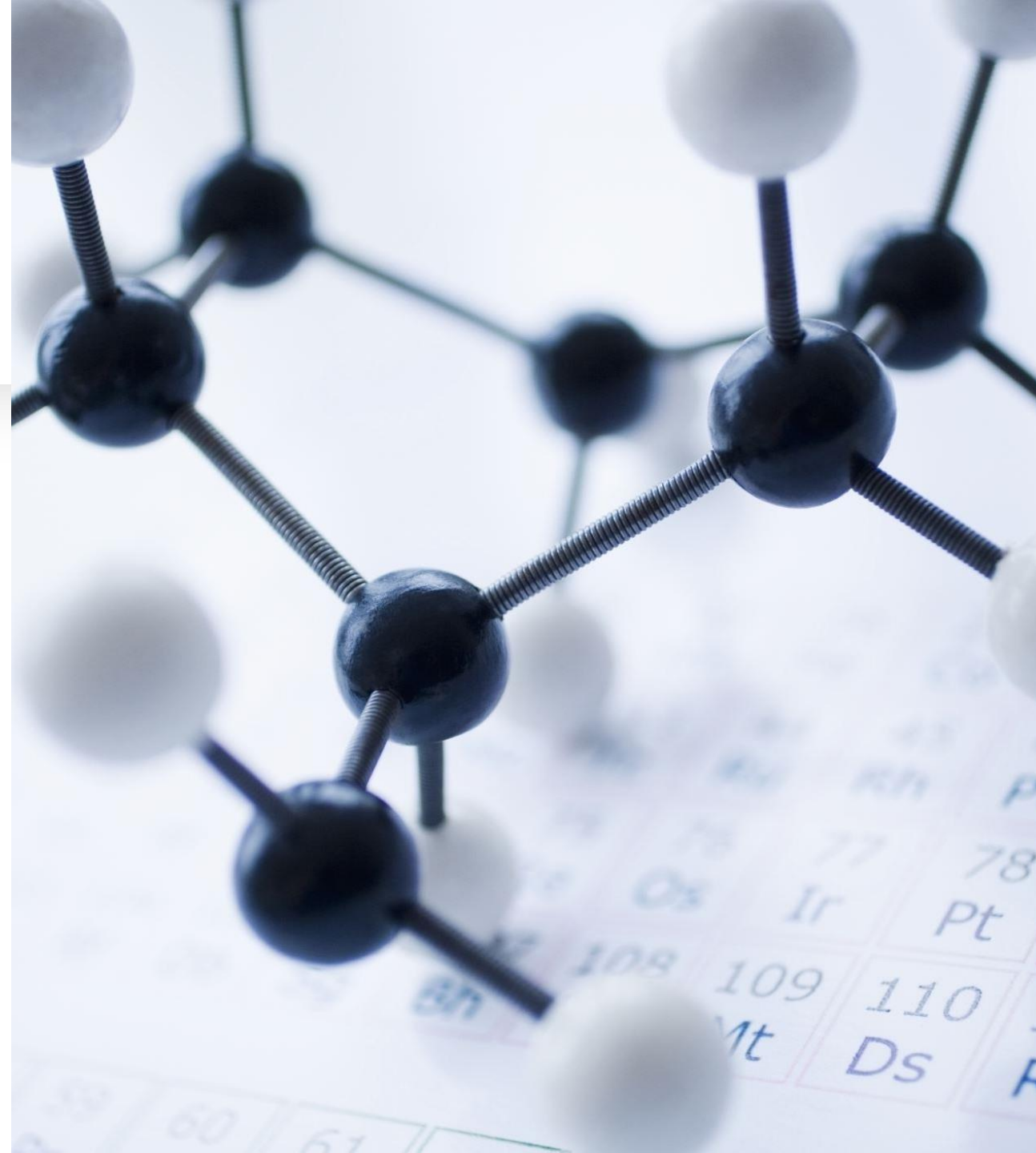
Goal: predict regulatory activity from
nucleotide sequence



Problem & Motivation

Why predict regulation from DNA sequence?

- Regulatory DNA controls gene expression
- Two canonical signals:
 - ATAC-seq: chromatin accessibility (broad regulatory activity)
 - CTCF: transcription factor binding (motif-driven, structural)
- Input: DNA window (A/C/G/T) → Output: probability of regulatory activity
- Need models that handle:
 - motifs + context
 - strand ambiguity (reverse complement)



Tasks & Data Setup

Two tasks, strict generalization protocol

- Genome: hg19
- ATAC: train on most chromosomes, validate on chr3–4
- CTCF: train chr4–22, validate chr2–3, test chr1 (held out)
- Why chromosome split?
 - prevents local genomic leakage
 - tests true generalization to unseen chromosomes





Dataset Construction

Peak-centered positives + improved negatives

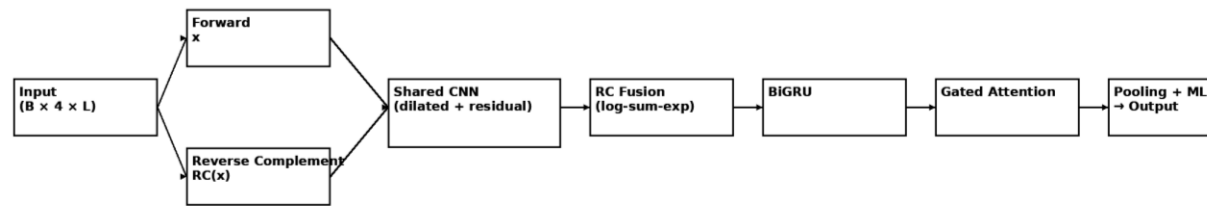
- Positives: midpoint-centered windows from peaks
 - Negatives are not random: we enforce realism
 - multiple negatives per positive
 - sampled far from any peak (min gap)
 - GC-matched to the corresponding positive
 - filter N-heavy windows
 - Benefit: prevents shortcut learning and forces learning real regulatory patterns
-

Baselines & Comparison Models

Controlled comparisons (same pipeline, same splits)

- Classical baselines:
 - Logistic regression
 - MLP
 - CNN baseline
- SOTA-inspired architectures (fair adaptations):
 - DeepSEA-style CNN
 - DanQ-style CNN + BiRNN
 - Basenji-style dilated CNN
- All trained/evaluated identically → architecture is the variable

Model Architecture



RC-dilated-CNN + BiGRU + Gated Attention

Pipeline:

- Input one-hot DNA
- Forward + reverse complement processed with shared CNN
- RC fusion (log-sum-exp)
- Dilated residual CNN blocks (longer receptive field)
- BiGRU (sequence dependency)
- Gated attention (focus/refine)
- Global pooling + MLP → prediction

Key idea: combine strong inductive biases for regulatory sequence modeling

Proposed Model (Architecture)

Training & Evaluation

Training details that mattered

- Loss: weighted BCEWithLogits (class imbalance)
- Optimizer: AdamW
- Schedule: OneCycleLR
- Stabilizers: mixed precision + EMA weights + early stopping
- Metrics: Accuracy, F1, AUROC, AUPRC
- Threshold chosen on validation to maximize F1

Results: ATAC (Harder Task)

ATAC is where the gap is largest

- CNN baseline strong but limited
- DanQ/Basemji-style improve
- Ours is best across all metrics (Ours $\approx$ 0.969 AUROC / 0.967 AUPRC)
- ~3 AUPRC points above strongest SOTA-inspired baseline

Interpretation: ATAC needs both motif + context + dependency modeling

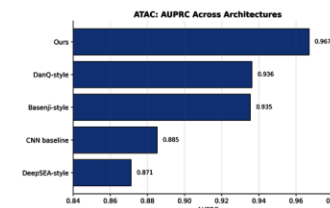
Results: CTCF (Easier, Motif- Driven)

- CTCF is high-performance across models — ours still leads
- Many models reach very high AUROC/AUPRC
- Ours remains best (≈ 0.995 AUROC/AUPRC)

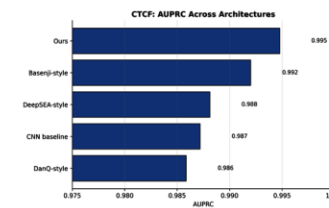
Interpretation: CTCF is more motif-dominant, but the combined architecture still helps

Results Summary

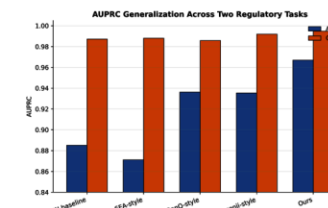
- ATAC shows largest gap (harder task)
- CTCF is easier overall, but model still leads
- Gain plot shows consistent improvement across metrics
- Multi-seed robustness supports stability



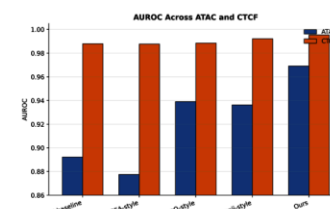
01_atac_auprc_comparison.png



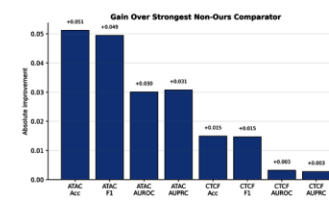
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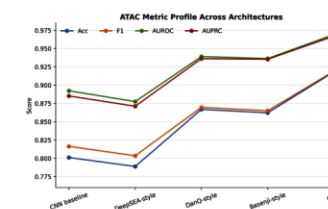
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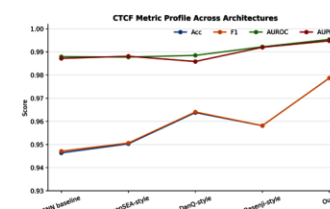
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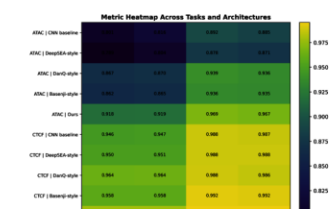
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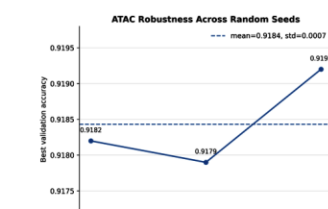
06_atac_metric_profile.png



07_ctcf_metric_profile.png



08_metric_heatmap.png



09_atac_multiseed_robustness.png



Robustness, Takeaways, Next Steps

Robustness & conclusions

- Multi-seed ATAC runs show stable validation accuracy (~0.918–0.919)
- Consistent wins on two distinct tasks
→ generalizable architecture

Takeaways

- RC handling + dilated context + recurrent modeling + attention = strong inductive bias
- Improved negative sampling matters for fair, realistic evaluation

Next steps (high-impact)

- Component ablations (RC, GRU, dilation, attention)
- Interpretability: saliency / in-silico mutagenesis / motif discovery
- More TFs / cell types, larger benchmarks